

Modern, Honest, and Open Multimodal Prediction of Post-Earnings-Call Volatility

(anonymized working draft for comments — please do not cite or circulate)

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Abstract

A decade of work predicts post-earnings-call stock volatility from the text and audio of quarterly conference calls, yet much of this literature compares against weak baselines and may reward models that memorise *which company* is speaking rather than *what* is said. We revisit the task with modern open-weight models, strong econometric baselines, leakage-proof evaluation, and an explicit identity-control suite. We build ECVOL, an open multimodal volatility benchmark over the FinCall-Surprise and MAEC corpora, providing three target families (level log-realised-volatility, change Δv , and HAR-residual), temporal and ticker-disjoint splits with a 30-trading-day embargo, SHA-256 data manifests, and a reproducible evaluation harness. Against persistence, EWMA, HAR-RV, GARCH(1,1), and ticker-fixed-effect baselines, we evaluate a ladder of frozen-representation content models (financial sentence embeddings, FinBERT, self-supervised and prosodic audio features) with light heads, under significance testing (Diebold–Mariano, cluster bootstrap, Holm correction) and identity controls (linear identity probes and same-ticker content shuffles). The evidence is consistent and negative: text embeddings encode *which company* is speaking (89.5% ticker-probe accuracy; within-ticker transcript shuffles leave performance unchanged); frozen audio content is inert beyond past volatility (its apparent gains survive even a global audio shuffle); fusion never significantly beats the best unimodal model; and LLM-extracted “auditable” semantic features fail a pre-registered human-validity audit ($\kappa \leq 0.222$ on every confirmatory field vs. a 0.6 gate, with prompt, quantization, and model-scale interventions all measurably null)—so, unlike prior work that consumes such features unvalidated, we do not scale them to the corpus. Under honest controls, apparent content signal in this task family is overwhelmingly company identity plus past volatility. [pending: A frozen-pipeline lookahead evaluation on post-LLM-cutoff 2025–2026 calls completes the diagnostic suite.] We release all code, configs, manifests, and result tables.

1 Introduction

Earnings conference calls are one of the few recurring events at which management speaks at length, semi-spontaneously, about a public company—in both *what* is said (the transcript) and *how* it is said (the audio). A line of research beginning with [Qin and Yang \(2019\)](#) and continuing through HTML ([Yang et al., 2020](#)), MAEC ([Li et al., 2020](#)), KeFVP ([Niu et al., 2023](#)), and ECC Analyzer ([Cao et al., 2024a](#)) claims that multimodal models over this data predict post-call stock volatility better than classical methods.

Two recent observations cast doubt on how much of that progress is real. First, [Yu et al. \(2024\)](#) (“Same Company, Same Signal”) show that transcript representations used across this literature predominantly encode *company identity* rather than content: same-ticker transcripts cluster tightly, and training-free baselines built on a company’s own past volatility beat all transcript-based models on their evaluation. Because volatility is strongly autocorrelated and company-specific, a model that learns “which company is this” inherits most of the apparent skill. Second, a 2025 survey of financial NLP ([Tatarinov et al., 2025](#)) finds that only $\sim 14\%$ of reported results reproduce exactly, attributing the gap to dead data links, API drift, and contamination. Together these suggest that the field’s leaderboard may be partly an artifact of identity leakage, weak baselines, and irreproducibility.

This paper reworks an abandoned 2024 study in this lineage from the ground up, with three goals. (i) **Modernise:** replace 2019-era components (GloVe, Praat-only acoustics, hand-rolled hierarchical transformers) with current open-weight models—financial sentence embeddings, FinBERT, self-supervised speech and prosodic features, and instruction-tuned LLMs with constrained decoding. (ii) **Open & reproduce:** use only openly licensed data and open-weight models, pin the environment, and release everything needed to rebuild the study, designing explicitly against the reproducibility crisis. (iii) **Evaluate honestly:** measure every model against strong econometric baselines under identity-aware controls and leakage-proof splits, so that any positive claim is attributable to content rather than identity.

Concretely, we ask four research questions:

RQ1 Does earnings-call *content* (text and/or audio) add predictive signal for post-call volatility beyond past-volatility persistence (HAR-RV, GARCH)?

RQ2 Is multimodal (audio + text) better than the best unimodal model, after controlling for past volatility?

RQ3 Do LLM-extracted *structured, auditable* features (guidance, hedging, evasiveness, ...) outperform opaque dense embeddings?

RQ4 Do any positive results survive (a) ticker-identity controls, (b) ticker-disjoint splits, and (c) evaluation on post-LLM-knowledge-cutoff calls?

The study is pre-registered so that no single outcome strands the work: a clean negative or mixed answer to RQ1/RQ4 is itself a publishable contribution. The paper’s framing is decided by the data at a pre-registered gate (§7), not retro-fitted to it. That gate has now been tested at the text, audio, and fusion stages and missed at each; it is provisionally resolved toward rigorous re-examination, with the post-cutoff lookahead experiment (in progress) providing final confirmation. This draft keeps a framing-neutral tone pending that last result.

Contributions.

1. **ECVOL, an open multimodal volatility benchmark** over FinCall-Surprise (primary) and MAEC (secondary): three target families (level v , change Δv , HAR-residual), temporal and ticker-disjoint splits with a 30-trading-day embargo, SHA-256 data manifests, and a reproducible harness.
2. **Honest baselines everywhere:** persistence, EWMA, HAR-RV, GARCH(1,1), and ticker-fixed-effect models reported for every horizon, split, and target variant—the comparison most prior work omits.
3. **A modern open-model feature ladder on consumer hardware:** financial embeddings, FinBERT, WavLM / emotion2vec+ / eGeMAPS audio features, fusion heads, and grammar-constrained LLM extraction—all runnable on a single 16–24 GB GPU.
4. **A diagnostic suite** for this task family: identity probes, transcript- and audio-shuffle controls, ticker-disjoint evaluation, a human-validity (κ) gate on LLM feature extraction—to our knowledge the first in this literature, and it fails—plus lookahead-bias tests on post-cutoff data and a gender-confound analysis.

2 Related Work

The benchmark lineage. Qin and Yang (2019) introduced text+audio volatility prediction and the 572-call EarningsCall dataset, establishing the 3/7/15/30-trading-day log-realised-volatility target family that the field still uses. MAEC (Li et al., 2020) scaled this to ~3,443 sentence-aligned multimodal calls over the S&P 1500. HTML (Yang et al., 2020) introduced a hierarchical transformer with multi-task learning across horizons; VolTAGE (Sawhney et al., 2020) added graph convolution over inter-stock relations with text/audio fusion; DialogueGAT (Sang and Bao, 2022) modelled speaker/dialogue structure with graph attention. KeFVP (Niu et al., 2023) combined knowledge-enhanced pre-training with a price-conditioned time-series module and is the standard SOTA reference. More recent work moves toward LLMs: ECC Analyzer (Cao et al., 2024a) uses hierarchical LLM summarisation with a RAG “question bank” and wav2vec2 features; RiskLabs (Cao et al., 2024b) fuses calls with news and market series; AT-FinGPT (Liu et al., 2025) is an audio-text LLM. Numeral-aware variants (NumHTML (Yang et al., 2022), NAM/ECNum (Chen et al., 2021a), GNAVol (Shi et al., 2023)) and Q&A-structure models (Ye et al., 2020) target specific aspects of call language. Closest to our structured-feature stage, recent work labels managerial evasion in call Q&A at scale via LLM consensus (Ma et al., 2026) and extracts KPIs with open-weight LLMs under human verification (Aavang et al., 2026); neither predicts a market outcome, and no prior system combines schema-constrained extraction, a human-agreement gate, and volatility prediction under identity controls.

Audio and prosody. Chen et al. (2025) report that physics-informed acoustic features explain up to 43.8% of 30-day realised-volatility variance, motivating an audio ladder; the FinAudio benchmark (Cao et al., 2025) cautions that audio-LLMs struggle on long financial audio, informing our preference for frozen pooled features over end-to-end audio-LLMs in v1.

Known pitfalls (the methodological core). Our design is organised around four documented failure modes. (1) *Ticker-identity leakage* (Yu et al., 2024) is the central threat to validity (§1); LLMs also deanonymise entity-neutered call transcripts with near-perfect accuracy (Lopez-Lira et al., 2025), so identity leaks even under masking. We respond with identity-robust targets (Δv , HAR-residual), a ticker-fixed-effect baseline, and an identity-control suite (§5). (2) *LLM lookahead bias*: a 2024-cutoff model “predicting” 2019–2021 outcomes it has effectively seen. A now-active literature offers diagnostics (Gao et al., 2025), cutoff-bounded pretraining (Yan et al., 2026; He et al., 2025), workflow benchmarks that find significant bias in open model families we use (Benhenda, 2026), and memorization filtering

(Roy and Roy, 2026); the nearest precedent validates an LLM disagreement measure post-cutoff on ECB press conferences (Collodel, 2025). None evaluates a frozen earnings-call pipeline on post-cutoff calls; we respond with exactly that, plus a prompt-masking ablation. (3) *Weak baselines*: much prior work omits persistence, GARCH(1,1), and HAR-RV (Corsi, 2009), which are hard to beat at long horizons; we compute these first and use persistence as the denominator of our headline metric (Gu et al., 2020). (4) *Gender bias in audio features* (Wang et al., 2024; Lin et al., 2024; Mathur et al., 2022); we analyse and report a gender confound rather than debias. We deliberately exclude agentic trading systems and time-series foundation models from v1 scope.

3 Data

3.1 Corpora

We build on two openly licensed multimodal earnings-call corpora and reserve a third, fresh corpus for a look-ahead test.

FinCall-Surprise (Shu et al., 2025) (primary; Apache-2.0) contains 2,688 calls spanning 2019–2021 with full MP3 audio, JSON transcripts, and slides. **MAEC** (Li et al., 2020) (secondary; CC-BY-SA-4.0) contains ~3,443 calls over 2015–2018 (S&P 1500) with sentence-aligned audio features and transcripts but no raw audio. A fresh **2025–2026 post-cutoff corpus** (≥ 200 calls) is acquired by released scripts for the lookahead experiment (§5); raw audio/transcripts from sources with unclear licenses are never redistributed (a *scripts-not-data* policy).

FinCall-Surprise ships without ticker/date metadata, which we reconstruct from transcript content and cross-check against price data; identity reconstruction resolves 2,496/2,688 calls (92.9% overall; 95.3% of the 2,499 earnings-type calls) and passes a 60/60 manual audit.

3.2 Price data and targets

Daily adjusted-close OHLCV is obtained from a free, no-account source and cross-checked against Tiingo on a random ticker sample (adjusted-return correlation > 0.999); delisted or unmatched tickers produce an explicit coverage report rather than being silently dropped. After price joining, FinCall-Surprise yields 2,424 calls with at least one target (97.1% join rate) and MAEC yields 2,578 (75.4%; a documented shortfall driven by 2015–2018 delistings), for a combined modelling set of 5,002 calls.

For each call we define returns $r_t = (P_t - P_{t-1})/P_{t-1}$ relative to day $t = 0$, the last trading day before the call’s information is public. Because $< 4\%$ of calls carry usable timestamps, we apply an after-hours rule (calls assumed to occur after market close; $t=1$ is the next trading session) and flag it. The primary target is log realised volatility over horizon $\tau \in \{3, 7, 15, 30\}$

trading days,

$$v_{\text{post}}(\tau) = \ln\left(\sqrt{\frac{1}{\tau} \sum_{t=1}^{\tau} (r_t - \bar{r})^2}\right), \quad (1)$$

with pre-call volatility $v_{\text{pre}}(\tau)$ defined over the τ sessions ending at day 0. We add two identity-robust variants: the **change** $\Delta v(\tau) = v_{\text{post}}(\tau) - v_{\text{pre}}(\tau)$, which subtracts the company’s own level, and the **HAR-residual** $v_{\text{post}}(\tau) - \widehat{\text{HAR}}(\tau)$, deviation from a train-only HAR-RV (Corsi, 2009) forecast. Insufficient post-call history and zero-variance windows yield NaN targets excluded with a reason code.

3.3 Leakage-proof splits

Three split families are generated once as committed CSVs and guarded by leakage assertions in CI. *Temporal*: train $<$ validation $<$ test in strict calendar order ($\sim 70/10/20$) with a **30-trading-day embargo** between segments so that no 30-day target window crosses a boundary—a constraint most prior work ignores. *Ticker-disjoint*: companies in validation/test never appear in training (grouped, seeded), the headline identity-control split. *Combined*: both constraints simultaneously, reported as the hardest robustness row. An information rule is codified and enforced: every feature function receives an `as_of` timestamp, and the harness fails any feature that reads price data after it. FinCall-Surprise spans COVID; per-year breakdowns are mandatory and a Feb–Apr 2020 exclusion is reported as an exploratory sensitivity.

4 Methods

We organise models as a ladder (Stages 0–5; Figure 1). Two invariants make the comparison honest. First, from Stage 2 upward every learned model includes past-volatility covariates (v_{pre} at all horizons plus the HAR-RV forecast) and is *also* reported with them ablated, baking the persistence control into every comparison. Second, given the small-data regime (10^3 – 10^4 samples) we use frozen representations with light heads (ridge, shallow MLP, LightGBM) and per-horizon models by default; fine-tuning is gated to Stage 6 and a shared multi-task head is an ablation, not the headline. Because the shallow-MLP heads proved seed-unstable (across-seed R_{OOS}^2 std up to ≈ 3.4 ; see Limitations), the consolidated grid (Table 5) uses one pre-registered deterministic ridge/structural canonical model per stage; MLP heads appear in the appendix only.

Stage 0 — econometric baselines. Persistence ($\hat{v} = v_{\text{pre}}$), EWMA (RiskMetrics $\lambda=0.94$), HAR-RV (OLS on daily/weekly/monthly realised-variance factors, Corsi, 2009), and GARCH(1,1) (per-call fit, `arch`). These establish the floor and the denominator of the headline metric.

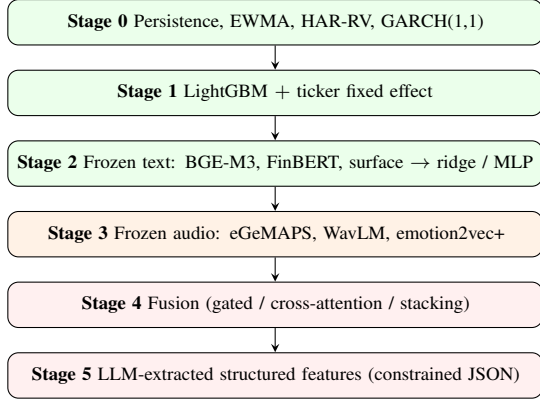


Figure 1: The modelling ladder. From Stage 2 up, every model also carries past-volatility covariates (and is reported with them ablated). Green = complete, orange = in progress, red = planned. [TODO: update shading as phases land].

Stage 1 — tabular identity baseline. LightGBM over past-volatility features, trivial metadata (transcript length, Q&A turn count where available), and a **ticker fixed effect**. This is the in-house “Same-Company-Same-Signal” control: it measures how much apparent multimodal performance is identity plus metadata.

Stage 2 — frozen text content. We segment each transcript into prepared-remarks vs. Q&A and speaker turns, then extract three feature blocks: BGE-M3 (BAAI, 2024) section-pooled sentence embeddings (1024-d), FinBERT (ProsusAI, 2019) sentiment aggregates per scope×role, and surface statistics. Heads are ridge (with the α grid selected on validation, *no* retrain on train+val) and a shallow MLP (1×256, L2 + early stop, 5 seeds). Features use fp32 with content-hash caching for bit-identical re-extraction.

Stage 3 — frozen audio content. Audio is QC’d and resampled to 16 kHz mono FLAC. We extract eGeMAPSv02 functionals (88-d, openSMILE (Eyben et al., 2010), CPU, interpretable), then WavLM-Large (Chen et al., 2021b) and emotion2vec+ (Ma et al., 2023) embeddings over 30 s non-overlapping windows, mean-pooled to a per-call vector (2,671 calls, all extracted on one consumer GPU with resumable content-hash caches), fed to the same heads. We use per-call pooling only (no per-sentence alignment), and report a gender-confound analysis.

Stage 4 — fusion. Gated fusion (per-modality train-fit PCA + norm gates → shallow MLP) and late-fusion stacking (a meta-ridge over the text, audio, and Stage-1 GBDT base predictions, fit on validation only) over frozen modality embeddings, to test whether multi-modal beats the best unimodal model (RQ2). Cross-attention is omitted: with one pooled vector per modality, attention over 2–3 tokens reduces to gated fusion. All fusion heads are <5M parameters.

Stage 5 — LLM-extracted structured features. Qwen2.5-7B-Instruct (Qwen Team, 2024) (4-bit) with grammar-constrained JSON decoding (JSON-schema grammar via `llama.cpp`; Outlines (dottxt-ai, 2023) on the GPU-server path) extracts auditable per-section features (guidance direction, hedging intensity, Q&A evasiveness, surprise mentions, analyst tone) against a frozen, human-signed-off schema and rubric, with YaRN-extended 65,536-token context so whole sections are never truncated. A pre-registered human-audit gate governs the stage: corpus-scale extraction is blocked until the model reaches $\kappa > 0.6$ against human rubric labels on 50 frozen train-only calls, extracted through the *same* engine and configuration as the corpus run (RQ3). As §6.6 reports, the gate failed and corpus extraction was not performed.

5 Experimental Protocol

Metrics. We report MSE on log-RV per horizon (literature comparability); R_{OOS}^2 vs. persistence (Gu et al., 2020; $1 - \sum (y - \hat{y})^2 / \sum (y - \hat{y}_{\text{persist}})^2$), the headline metric, which is not gameable by identity memorisation the way raw MSE is; MAE; and cross-sectional Spearman rank correlation per calendar quarter. All metrics are reported on all three targets (v , Δv , HAR-residual) × all splits.

Statistical rigor. Every learned model is run with ≥ 5 seeds (mean $\pm$ std). We use Diebold–Mariano tests (Diebold and Mariano, 1995) (HLN small-sample correction) on per-call squared-error differentials vs. persistence and vs. the best preceding stage; cluster-bootstrap confidence intervals (1,000 resamples) clustered by ticker and by calendar quarter; and Holm-corrected p -values across horizons for confirmatory claims. Confirmatory vs. exploratory analyses are labelled in advance: confirmatory analyses are Stage- k vs. Stage-1 DM tests on Δv at each τ , Stage 4 vs. best unimodal, Stage 5 vs. Stage 2, and the temporal-vs-ticker-disjoint gap per stage; all per-year/regime, COVID-exclusion, prepared-vs-Q&A, call-length, and gender-confound analyses are exploratory.

Identity-control suite. The signature diagnostic of this study probes whether content models read content or identity, via four controls. (1) *Ticker-only model*: predict the per-ticker target mean (global mean for unseen tickers); if a content model matches it on level- v , that model reads identity. (2) *Same-ticker transcript shuffle*: replace each test call’s transcript with a different call from the *same* ticker; if performance is unchanged the model reads identity, not content, whereas a *global* shuffle (any other call) isolates pure overfitting. (3) *Identity probe*: a linear probe predicting the ticker from frozen call embeddings; high accuracy demonstrates identity is

encoded. (4) *Ticker-disjoint split* results reported beside temporal-split results in every table.

Lookahead-bias experiments. [pending: Phase 7 — in progress: a 50-call Q2-2026 capture pilot (replay audio, native identity, exact call timestamps) is running ahead of the full acquisition.] The entire frozen pipeline—no retraining, no post-hoc threshold changes (rule fixed in advance)—will be evaluated on the 2025–2026 post-cutoff corpus; disproportionate degradation of Stage-5 LLM features relative to other stages would be contamination evidence. A masking ablation (company names, tickers, dates removed from Stage-5 prompts) will estimate identity-mediated leakage.

Reproducibility. All results regenerate byte-identically from committed CSV artifacts via `ecvol` report, guarded by a CI test. Every data file carries a SHA-256 manifest entry, and every run artifact is keyed by config hash plus git SHA.

6 Results

We report results factually and defer the headline framing to §7. Curated headline tables appear here on the identity-robust Δv target for FinCall-Surprise; the complete grid (both corpora, level- v and Δv , R^2_{OOS} and MSE, all splits and horizons) is in Appendix A. Stars denote Diebold–Mariano significance ($p < 0.05$): in Table 1 vs. persistence, and in Tables 2 and 4 vs. the Stage-1 ticker-FE GBDT; in the consolidated grid (Table 5) markers are Holm-corrected across horizons and signed (* better / † worse than Stage-1).

6.1 Baselines (Stages 0–1)

Table 1 shows the econometric floor. HAR-RV is strong at short and medium horizons on both splits, confirming RQ1’s premise that past-volatility baselines are hard to beat. The one striking failure is the FinCall *temporal* split at $\tau=30$, where every model falls below persistence (HAR-RV $R^2_{\text{OOS}} = -0.287$): the training window is COVID-inflated (Feb–May 2020) while the test window is calm late 2021, so a long-horizon level forecast trained on the crisis regime is mis-calibrated. The same models pass comfortably on the ticker-disjoint split (HAR-RV $R^2_{\text{OOS}} = +0.229$ at $\tau=30$) and on MAEC, so we treat this as a documented regime exception rather than a target bug; per-year breakdowns are mandatory in the appendix.

6.2 Text content (Stage 2)

Table 2 shows frozen-text heads against the Stage-1 baseline on Δv . The result is a clean negative: *text content does not cleanly beat the identity/past-volatility*

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
<i>Temporal split (n=481)</i>				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.387*	0.362*	0.145*	-0.219*
HAR-RV	0.464*	0.410*	0.207*	-0.287*
GARCH(1,1)	0.265*	0.193*	-0.140*	-0.541*
GBDT (ticker FE)	0.344*	0.248*	-0.057	-0.904*
<i>Ticker-disjoint split (n=484)</i>				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.357*	0.199*	-0.009	0.046
HAR-RV	0.495*	0.357*	0.211*	0.229*
GARCH(1,1)	0.356*	0.229*	0.088	0.209*
GBDT (ticker FE)	0.407*	0.275*	0.089	0.108

Table 1: Baselines. R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). HAR-RV is strong except on the COVID-regime temporal $\tau=30$ cell. *: Diebold–Mariano significant vs. persistence ($p < 0.05$).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
<i>Temporal split (n=481)</i>				
Ridge (text)	0.028*	0.030*	0.015	-0.144*
Ridge (past-vol)	0.415*	-1.053	-2.858	-0.667
Ridge (text+vol)	-1.678	-3.076	-3.586	-0.468
MLP (text)	-0.108*	-0.175*	-0.265*	-1.042
<i>Ticker-disjoint split (n=484)</i>				
Ridge (text)	-0.044*	-0.026*	0.008	0.171
Ridge (past-vol)	0.451	0.306	0.191*	0.224*
Ridge (text+vol)	0.223*	0.129*	0.094	0.236
MLP (past-vol)	0.492*	0.363*	0.230*	0.249*

Table 2: Text content (Stage 2). R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). The text-only head does not separate from the past-volatility head; adding text to past-vol does not help. *: Diebold–Mariano significant vs. the Stage-1 ticker-FE GBDT ($p < 0.05$). Full head set in Appendix A.

baselines. On the FinCall temporal split the text ridge head is near zero and collapses at $\tau=30$; the MLP heads overfit badly. On the ticker-disjoint split the text ridge head is weak (R^2_{OOS} between -0.04 and $+0.17$) while the past-vol-only head is consistently strong ($R^2_{\text{OOS}} \approx 0.19$ – 0.45), so any apparent text gain is not separable from what past volatility already provides. MAEC shows the same pattern at smaller magnitude (Appendix A).

6.3 Identity controls (Stage 2 diagnostic)

Table 3 explains the negative result. A linear probe recovers the ticker from frozen call embeddings at 89.5% on FinCall ($319\times$ chance) and 53.9% on MAEC ($487\times$ chance): identity is heavily encoded. The transcript shuffle is decisive—replacing each test call’s transcript with a *different call from the same ticker* barely changes R^2_{OOS} , whereas a global shuffle degrades it. The heads are reading *which company* is speaking, not *what* is said. This is the in-corpus confirmation of Yu et al. (2024)

Corpus	Identity probe (linear)	
	Accuracy	$\times$ chance
FinCall ($n=2656$, 357 tickers)	89.5%	319 $\times$
MAEC ($n=3132$, 902 tickers)	53.9%	487 $\times$

Condition	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Real transcript	0.028	0.030	0.015	-0.144
Within-ticker shuffle	0.017	0.020	0.006	-0.142
Global shuffle	-0.099	-0.121	-0.115	-0.266

Table 3: Identity controls. Top: a linear probe predicts the ticker from frozen BGE-M3 call embeddings far above chance, demonstrating identity is heavily encoded. Bottom: R^2_{OOS} of the text ridge head on the Δv target (FinCall, temporal split, test, $n=481$). Replacing each test call’s transcript with a *different call from the same ticker* (within-ticker shuffle) barely changes performance, whereas a global shuffle degrades it—evidence the head reads ticker identity rather than call content.

and the reason RQ4 is central.

6.4 Audio content (Stage 3)

Table 4 extends the same design to audio. Audio-only heads (eGeMAPS, WavLM-Large, emotion2vec+, ridge or MLP) sit at R^2_{OOS} zero on every split and horizon (-0.06 to $+0.03$): frozen audio content alone carries no measurable Δv signal. Adding past-volatility covariates reproduces the Stage-2 pattern—the only DM-significant wins over Stage-1 are past-vol-augmented heads on the ticker-disjoint split. The audio-shuffle control (bottom block) is decisive: replacing each test call’s audio with a random *other company’s* call leaves most of those gains intact ($\tau=3$: $0.337 \rightarrow 0.318$; $\tau=30$: $0.212 \rightarrow 0.100$), so the audio features are inert and past volatility carries the signal. Identity is nonetheless encoded: a linear probe recovers the ticker from per-call WavLM embeddings at 76.3% ($271\times$ chance; emotion2vec+ 29.4%, $105\times$). An exploratory gender-confound check (eGeMAPS-F0 dominant-speaker proxy, 100% coverage) finds near-identical per-group test MSE (0.241 low-pitch vs. 0.242 high-pitch) and a 0.03 correlation between speaker F0 and squared error: with inert audio features, no large gender disparity is detectable (see the Ethics Statement for caveats).

6.5 Fusion and the consolidated grid (Stage 4)

Gated fusion and late-fusion stacking over all frozen modalities answer RQ2 in the negative: *multimodal is not better than the best unimodal model*. Only one of 48 Δv test cells beats Stages 1–3 DM-significantly (gated fusion, temporal $\tau=7$, $R^2_{\text{OOS}} = +0.019$ —negligible) and the best fusion cell overall (stacking, ticker-disjoint, $\tau=3$, $R^2_{\text{OOS}}=0.436$) is not significant vs. Stage-1

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
<i>Temporal split ($n=481$)</i>				
Ridge (eGeMAPS)	-0.000*	-0.007*	-0.015	-0.007*
Ridge (WavLM)	-0.005*	-0.007*	-0.045	-0.031*
Ridge (emotion2vec+)	-0.018*	-0.028*	-0.056	-0.060*
Ridge (WavLM+past-vol)	0.265	-1.026	-2.032	-0.561
<i>Ticker-disjoint split ($n=484$)</i>				
Ridge (eGeMAPS)	-0.011*	-0.013*	-0.002	0.002
Ridge (WavLM)	-0.038*	-0.035*	-0.027*	0.024
Ridge (emotion2vec+)	-0.002*	-0.003*	0.011	0.002
Ridge (WavLM+past-vol)	0.337*	0.171*	0.111	0.212*
<i>Audio-shuffle control of WavLM+past-vol (ticker-disjoint)</i>				
Real audio	0.337	0.171	0.111	0.212
Within-ticker shuffle	0.317	0.144	0.097	0.170
Global shuffle	0.318	0.115	0.093	0.100

Table 4: Audio content (Stage 3). R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). Audio-only heads sit at zero everywhere; the apparent gains of the past-vol-augmented head survive replacing every test call’s audio with a *random other company’s* call (global shuffle, bottom block), so past volatility, not audio content, carries them. *: Diebold–Mariano significant vs. the Stage-1 ticker-FE GBDT ($p < 0.05$; read direction from the value). Full head set in Appendix A.

($p=0.13$). Table 5 consolidates the ladder with one pre-registered canonical model per stage and Holm-corrected confirmatory tests. Two facts summarise it. On the regime-honest *temporal* split, no content stage beats the econometric floor at any horizon; where a content model is Holm-significantly better than Stage-1 (fusion at $\tau=30$) it merely beats a collapsed GBDT while remaining below persistence. On the *ticker-disjoint* split the favorable cells are exactly the past-vol-augmented ones that the shuffle controls (Tables 3 and 4) show to be identity/past-vol-confounded. The per-year breakdown sharpens the regime point: the FinCall temporal test year is entirely 2021, and there every content model underperforms persistence on level- v (HAR -0.29 , GBDT -0.82 , text -1.09 , audio -1.66 , fusion -1.02).

6.6 LLM structured features (Stage 5): the extraction fails its audit

Stage 5 was pre-registered with a validity gate: corpus-scale extraction of the structured features is blocked until the model’s labels reach $\kappa > 0.6$ against human rubric labels on a frozen 50-call train-only audit sample, extracted through the *same* engine and configuration as the corpus run would use. The gate failed decisively (Table 6): the three confirmatory fields score κ of 0.165 (guidance direction), -0.050 (hedging intensity—worse than chance), and 0.222 (surprise mentions), with call-level-bootstrap 95% upper bounds all far below 0.6. Every output was 100% schema-valid, so this is precisely the gap between *form* and *content* validity: constrained decoding guarantees parseable JSON, not meaningful

Stage (canonical model)	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Field	Role	κ	95% CI
<i>Temporal split (n=481)</i>					Guidance direction	confirmatory	0.165	[0.064, 0.271]
S0 HAR-RV	0.464	0.410	0.207	-0.287	Hedging intensity	confirmatory	-0.050	[-0.124, 0.033]
S1 GBDT (ticker FE)	0.344	0.248	-0.057	-0.904	Surprise mentions	confirmatory	0.222	[0.054, 0.378]
S2 Text (ridge+vol)	-1.678	-3.076	-3.586	-0.468	Q&A evasiveness	reported	0.108	—
S3 Audio (ridge+vol)	0.214	-1.099	-3.706	-0.548	Analyst tone	reported	-0.040	—
S4 Fusion (stack)	0.232	-0.326	-1.828	-0.010*				
<i>Ticker-disjoint split (n=484)</i>								
S0 HAR-RV	0.495	0.357	0.211	0.229				
S1 GBDT (ticker FE)	0.407	0.275	0.089	0.108				
S2 Text (ridge+vol)	0.223†	0.129†	0.094	0.236				
S3 Audio (ridge+vol)	0.304†	0.147†	0.044	0.209				
S4 Fusion (stack)	0.436	0.292	0.133	0.244				

Table 5: Consolidated grid (Stages 0–4), pre-registered canonical model per stage. R_{OOS}^2 vs. persistence on the Δv target, FinCall-Surprise (test); persistence is 0 by construction. Significance is the Holm-corrected confirmatory family (Stage- k vs. Stage-1 DM across horizons): * = significantly better than Stage-1, † = significantly worse ($p < 0.05$). No content stage beats the econometric floor on the temporal split (the fusion $\tau=30$ star beats only the collapsed GBDT, not persistence); the ticker-disjoint gains are past-vol/identity-confounded (Table 4, bottom block). Canonical models are deterministic ridge/structural heads; seed-unstable MLP heads are reported in Appendix A only.

Table 6: LLM extraction validity (Stage 5). Cohen’s κ between constrained-decoded Qwen2.5-7B-Instruct outputs and a human rater on the frozen 50-call audit sample (97 call-section rows; Q&A-only fields $n=50$). The pre-registered gate requires $\kappa > 0.6$ on the confirmatory core before corpus-scale extraction; every confirmatory field’s 95% upper bound sits far below it, so the failure is not a small-sample accident. All outputs were 100% schema-valid—constrained decoding guarantees *form*, not *content* validity. Corpus-scale extraction was therefore not run (single rater; inter-annotator agreement pending, see §6).

unit going forward is model κ relative to human–human κ (see Limitations).

6.7 Post-cutoff lookahead (pending)

[pending: Phase 7: the frozen-pipeline evaluation on 2025–2026 post-cutoff calls. A 50-call capture pilot (replay audio with native identity and exact timestamps) is in progress.]

measurements.

Four controls make the diagnosis actionable, and all four capability levers are null. Sharpened prompt anchors move κ by at most +0.04; near-lossless Q8_0 quantization (vs. 4-bit) and a $2\times$ larger model (Qwen2.5-14B) both produce differences whose bootstrap CIs all span zero (the 14B also fails the gate on its own terms); the single measurable effect in the whole diagnostic set is few-shot exemplars of the rater’s own labels, which lift hedging from below chance to $\kappa \approx 0.09$ (+0.092 [+0.029, +0.181])—a fifth of the distance to the bar, on exactly the field whose written anchor omits a baseline. The failure modes are structural rather than noisy: the model rates 47 of 50 Q&A sections exactly “2” on analyst tone (near-constant output), its hedging ratings are statistically independent of the human’s, and it misses 26 of 36 human-positive surprise rows. Per the pre-registered rule, corpus-scale extraction was therefore not run: there is no Result Table 5, and RQ3 is answered at the validity stage—the “auditable semantics” this sub-literature consumes did not survive their audit. Notably, the closest prior system (Cao et al., 2024a) reports a 27.7% MSE reduction from LLM-extracted semantics with no human validation of extraction quality at all, and the highest published agreement figure in this space (Ma et al., 2026) is inter-LLM, not human, agreement. Two caveats travel with this finding: the gate scored a single careful rater (inter-annotator agreement pending), and the 0.6 bar’s implicit human–human ceiling was never measured—subjective ordinal annotation routinely sits at human–human κ of 0.4–0.6, so the honest reporting

7 Discussion

What the evidence shows. On the completed ladder (Stages 0–5), five statements hold. (i) Strong econometric baselines are hard to beat, and the headline metric (R_{OOS}^2 vs. persistence) exposes a long-horizon regime failure (FinCall temporal $\tau=30$) that raw MSE would obscure. (ii) Frozen text content does not cleanly add signal over past-volatility and identity baselines on the identity-robust Δv target, and the reason is measurable: call embeddings encode ticker identity (89.5% probe accuracy on FinCall), and within-ticker transcript shuffles leave performance essentially unchanged—an in-corpus replication of the “Same Company, Same Signal” effect (Yu et al., 2024). (iii) Frozen audio content is inert: audio-only heads sit at R_{OOS}^2 zero everywhere, the apparent gains of past-vol-augmented audio heads survive a global audio shuffle, and WavLM embeddings still encode the speaker’s company at 76.3% probe accuracy—identity without predictive content. (iv) Fusion does not rescue the modalities it fuses: multimodal never significantly beats the best unimodal model (RQ2 negative). (v) LLM-extracted “auditable” semantic features fail their own audit: constrained decoding delivers 100% schema-valid output whose content agrees with a human rater at $\kappa \leq 0.222$ on every confirmatory field, far below the pre-registered 0.6 gate, with capability levers (prompt anchors, quantization, $2\times$ scale) all measurably null. Together these form one coherent finding: *in*

this task family, what looks like content signal is overwhelmingly company identity plus past volatility, and the semantic features the recent literature consumes are not validated measurements.

Engaging the strongest positive claims. The strongest recent positive claim—up to 43.8% of 30-day realised-volatility variance from text+acoustic features (Chen et al., 2025)—reports no identity controls. Prosody was the modality most plausibly carrying content orthogonal to identity; our modern audio ladder (WavLM, emotion2vec+, eGeMAPS) finds it inert once past volatility is controlled—albeit on the corpus’s 40 kbps source audio (see Limitations)—and our shuffle controls predict such gains would not survive a within-ticker (let alone global) shuffle—a check we encourage and that our released suite makes cheap. Likewise, ECC Analyzer (Cao et al., 2024a) reports large MSE reductions from LLM-extracted semantics with no human validation of extraction quality; our Stage-5 gate is exactly that validation, and it failed for every confirmatory field.

The framing gate (pre-registered; provisionally resolved). The paper’s spine is decided at a pre-registered gate, not retro-fitted. *Path A (positive showcase)* is adopted only if content-bearing models beat the Stage-0/1 floor on Δv or HAR-residual with DM-significant improvements that survive the identity-control suite on at least two of four horizons. *Path B (rigorous re-examination)* is adopted otherwise. The Path-A criterion has now been tested and missed at Stage 2 (text), Stage 3 (audio), and Stage 4 (fusion), and Stage 5 was blocked at validity; the gate is therefore *provisionally resolved to Path B* (recorded in the project decision log), with final confirmation deferred to the post-cutoff lookahead test (Phase 7). This draft remains framing-neutral in tone pending that last experiment. **[TODO: Finalise the headline once the Phase-7 lookahead evaluation lands.]**

Instrument, not model. The Stage-5 failure decomposes informatively (§6.6). Part is instrument defect: the rubric’s “hedging recurs” anchor is trivially true of any 25k-character Q&A section, so the model’s constant “moderate” is arguably rubric-compliant while the human applied an unstated relative baseline—an ambiguity no model scale fixes, and consistent with the one non-null control (rater-baseline exemplars helping exactly that field). Part is genuine non-discrimination (near-constant analyst-tone output). And part is structural to the gate itself: $\kappa > 0.6$ against a single unvalidated rater has an unmeasured human–human ceiling and subjective ordinal annotation routinely ceilings at 0.4–0.6. The constructive programme is therefore behaviourally anchored, coarser rubrics plus a measured annotator ceiling—not larger models.

Why the contribution holds either way. Independent of the gate, ECVOL contributes an open, identity-controlled, leakage-proof multimodal benchmark with honest baselines and a reusable diagnostic suite—identity probes, shuffle controls, and an extraction-validity gate—precisely the apparatus the field has lacked. A negative or mixed answer to RQ1–RQ4 is a publishable result that the existing literature does not provide. Beyond the lookahead test, the natural next step is data rather than models: a forward-collected corpus with native identity and exact call timestamps (a capture pilot is underway) removes the reconstruction and after-hours-assumption caveats that this generation of corpora imposes.

8 Conclusion

We revisit multimodal earnings-call volatility prediction with modern open-weight models, strong econometric baselines, leakage-proof splits, and an explicit identity-control suite, released as the ECVOL benchmark. Across the completed ladder the evidence is consistent and negative: frozen text content reads company identity rather than call content (89.5% ticker-probe accuracy; within-ticker shuffles leave performance unchanged); frozen audio content is inert beyond past volatility (its apparent gains survive a global audio shuffle); fusion never significantly beats the best unimodal model; and LLM-extracted “auditable” semantic features fail their pre-registered human-validity audit ($\kappa \leq 0.222$ vs. a 0.6 gate), so were never scaled to the corpus. The pre-registered framing gate is provisionally resolved to rigorous re-examination: what the leaderboard literature measures as content signal is, under honest controls, overwhelmingly company identity plus past volatility—and the semantic features it increasingly consumes are not validated measurements. The post-cutoff lookahead evaluation (in progress) completes the diagnostic suite. **[TODO: Fold in the Phase-7 lookahead result and finalise the verdict.]**

Limitations

Lookahead still pending. The Phase-7 frozen-pipeline evaluation on post-cutoff 2025–2026 calls and the prompt-masking ablation are in progress; until they land, LLM-adjacent results on 2019–2021 calls carry contamination risk by default, and the framing-gate resolution remains provisional. **MLP-head instability.** The shallow-MLP heads are seed-unstable (across-seed R^2_{OOS} std up to ≈ 3.4), which is why the consolidated grid uses deterministic ridge/structural canonicals; MLP cells in the appendix should be read as seed means with that caveat. **The past-vol covariate block is not HAR.** The learned $[v_{\text{pre}}, \text{RV}_{1/5/22}]$ ridge used inside Stage-2–4 heads overfits the COVID regime shift on the temporal split ($R^2_{\text{OOS}} - 2.0$ at $\tau=15$) where the rigid

OLS HAR-RV stays robust (+0.21); the two are labelled distinctly throughout, and the gap is itself a small finding about structural vs. unconstrained conditioning under regime shift. **Short-horizon R_{OOS}^2 inflation.** The persistence denominator is noisiest at $\tau=3$ (its MSE is $\sim 2.8\times$ the $\tau=7$ one), so short-horizon R_{OOS}^2 gains are easier to show and should be read accordingly. **Times-tamps.** Fewer than 4% of calls carry usable times-tamps, so we apply a flagged after-hours rule; mis-dated calls add noise to short-horizon targets. **Regime sensitivity.** FinCall-Surprise spans COVID; the temporal $\tau=30$ baseline failure shows long-horizon level forecasts are regime-sensitive, which is why we emphasise the identity-robust Δv target and report per-year breakdowns. **MAEC coverage.** The 75.4% price-join rate (2015–2018 delistings) makes MAEC a secondary, comparability corpus; it also ships no raw audio, so the audio ladder is FinCall-only. **Source audio quality.** Every FinCall audio file is a uniform 40 kbps MP3 transcode (eight source sample rates; 14 calls at telephone-grade 8 kHz), so aggressive lossy compression underlies all acoustic features. The audio-inert finding is therefore established on compression-damaged audio and may not transfer to studio-quality recordings—a caveat that applies equally to prior positive claims built on the same corpora. **No diarisation in v1.** Audio uses per-call pooling, not speaker-resolved features; the gender analysis correspondingly uses a dominant-speaker F0 proxy. **The κ gate scored a single rater.** Inter-annotator agreement is pending, and the 0.6 bar’s implicit human–human ceiling was never measured; the extraction-failure claim is robust (all confirmatory 95% upper bounds $\ll 0.6$) but whether the fault lies with the model or the rubric awaits a second rater. **No corpus-scale LLM features exist.** Blocking extraction at the validity gate means RQ3’s predictive half (do valid structured features beat embeddings?) is unanswered, not answered negatively.

Ethics Statement

Data and licensing. We use only openly licensed corpora (FinCall-Surprise, Apache-2.0; MAEC, CC-BY-SA-4.0) and open-weight models. Derived numeric features inherit and explicitly state their upstream license; raw audio/transcripts from unclear-license sources are never redistributed (*scripts-not-data*). **Gender confound.** Audio features encode speaker gender and female executives are underrepresented in these corpora, which can produce error disparities (Wang et al., 2024; Lin et al., 2024; Mathur et al., 2022); we analyse and report a gender-confound diagnostic rather than silently propagate the bias. **Intended use.** This is a forecasting-research benchmark, not investment advice; predictions are noisy and regime-sensitive, and should not be used for automated trading. **Reproducibility as an ethical commitment.** Given the documented reproducibility crisis in financial NLP (Tatarinov et al., 2025), we pin the environment, checksum every data file, and regen-

erate all result tables byte-identically from committed artifacts.

References

- Rasmus T. Aavang, Rasmus Tjalk-Bøggild, Alexandre Iolov, Giovanni Rizzi, Mike Zhang, and Johannes Bjerva. Effective performance measurement: Challenges and opportunities in KPI extraction from earnings calls. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Industry Track)*, 2026. URL <https://arxiv.org/abs/2605.03147>.
- BAAI. BGE-M3 sentence embeddings. Hugging Face (MIT), 2024. URL <https://huggingface.co/BAAI/bge-m3>.
- Mostapha Benhenda. Look-ahead-bench: a standardized benchmark of look-ahead bias in point-in-time LLMs for finance. arXiv:2601.13770, 2026. URL <https://arxiv.org/abs/2601.13770>.
- Yupeng Cao et al. ECC analyzer: Extracting trading signal from earnings conference calls using large language models for stock volatility prediction. In *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF)*, 2024a. URL <https://arxiv.org/abs/2404.18470>.
- Yupeng Cao et al. RiskLabs: Predicting financial risk using large language model based on multimodal and multi-sources data. arXiv:2404.07452, 2024b. URL <https://arxiv.org/abs/2404.07452>.
- Yupeng Cao et al. FinAudio: A benchmark for audio large language models in financial applications. arXiv:2503.20990, 2025. URL <https://arxiv.org/abs/2503.20990>.
- Chen et al. Distilling numeral information for volatility forecasting. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, 2021a. doi: 10.1145/3459637.3482089.
- Sanyuan Chen et al. WavLM: Large-scale self-supervised pre-training for full stack speech processing. Hugging Face (MIT): microsoft/wavlm-large, 2021b. URL <https://huggingface.co/microsoft/wavlm-large>.
- Xiaoliang Chen et al. The sound of risk: A multimodal physics-informed acoustic model for forecasting market volatility and enhancing market interpretability. arXiv:2508.18653, 2025. URL <https://arxiv.org/abs/2508.18653>.
- Umberto Collodel. Interpreting the interpreter: Can we model post-ECB conferences volatility with LLM agents? arXiv:2508.13635, 2025. URL <https://arxiv.org/abs/2508.13635>.

- Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 2009. doi: 10.1093/jfinc/ebp001.
- Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 1995. doi: 10.1080/07350015.1995.10524599.
- dottxt-ai. Outlines: Structured generation / constrained decoding. github.com/dottxt-ai/outlines, 2023. URL <https://github.com/dottxt-ai/outlines>.
- Florian Eyben, Martin Wöllmer, and Björn Schuller. openSMILE / eGeMAPS acoustic feature extraction. github.com/audeering/opensmile-python, 2010. URL <https://github.com/audeering/opensmile-python>.
- Zhenyu Gao, Wenxi Jiang, and Yutong Yan. Detecting lookahead bias in LLM forecasts. arXiv:2512.23847, 2025. URL <https://arxiv.org/abs/2512.23847>.
- Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 2020. doi: 10.1093/rfs/hhaa009.
- Songrun He, Linying Lv, Asaf Manela, and Jimmy Wu. Chronologically consistent large language models. arXiv:2502.21206, 2025. URL <https://arxiv.org/abs/2502.21206>.
- Jiazheng Li, Linyi Yang, Barry Smyth, and Ruihai Dong. MAEC: A multimodal aligned earnings conference call dataset for financial risk prediction. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, 2020. doi: 10.1145/3340531.3412879.
- Yi-Cheng Lin et al. Emo-bias: A large scale evaluation of social bias on speech emotion recognition. arXiv:2406.05065, 2024. URL <https://arxiv.org/abs/2406.05065>.
- Y. Liu et al. AT-FinGPT: Financial risk prediction via an audio-text large language model. *Finance Research Letters*, 2025. URL <https://www.sciencedirect.com/science/article/abs/pii/S1544612325002314>.
- Alejandro Lopez-Lira, Yuehua Tang, and Mingyin Zhu. The memorization problem: Can we trust LLMs’ economic forecasts? arXiv:2504.14765, 2025. URL <https://arxiv.org/abs/2504.14765>.
- Shijian Ma, Yan Lin, and Yi Yang. EvasionBench: A large-scale benchmark for detecting managerial evasion in earnings call Q&A. arXiv:2601.09142, 2026. URL <https://arxiv.org/abs/2601.09142>.
- Ziyang Ma et al. emotion2vec: Self-supervised pre-training for speech emotion representation. Hugging Face: emotion2vec/emotion2vec_plus_large, 2023. URL https://huggingface.co/emotion2vec/emotion2vec_plus_large.
- Puneet Mathur et al. DocFin: Multimodal financial prediction and bias mitigation using semi-structured documents. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL <https://aclanthology.org/2022.findings-emnlp.139/>.
- Hao Niu et al. KeFVP: Knowledge-enhanced financial volatility prediction. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. URL <https://aclanthology.org/2023.findings-emnlp.770/>.
- ProsusAI. FinBERT: Financial sentiment analysis with bert. Hugging Face, 2019. URL <https://huggingface.co/ProsusAI/finbert>.
- Yu Qin and Yi Yang. What you say and how you say it matters: Predicting stock volatility using verbal and vocal cues. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019. URL <https://aclanthology.org/P19-1038/>.
- Qwen Team. Qwen2.5-7B-Instruct. Hugging Face (Apache-2.0), 2024. URL <https://huggingface.co/Qwen/Qwen2.5-7B-Instruct>.
- Anisha Roy and Dip Roy. MemGuard-Alpha: Detecting and filtering memorization-contaminated signals in LLM-based financial forecasting via membership inference and cross-model disagreement. arXiv:2603.26797, 2026. URL <https://arxiv.org/abs/2603.26797>.
- Yunxin Sang and Yang Bao. DialogueGAT: A graph attention network for financial risk prediction by modeling the dialogues in earnings conference calls. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL <https://aclanthology.org/2022.findings-emnlp.117/>.
- Ramit Sawhney, Piyush Khanna, Arshiya Aggarwal, Taru Jain, Puneet Mathur, and Rajiv Ratn Shah. VolTAGE: Volatility forecasting via text audio fusion with graph convolution networks for earnings calls. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020. URL <https://aclanthology.org/2020.emnlp-main.643/>.
- Shi et al. Enhancing volatility forecasting in financial markets: A general numeral attachment dataset for understanding earnings calls.

In *Proceedings of IJCNLP-AACL 2023*, 2023. URL <https://aclanthology.org/2023.ijcnlp-short.5/>.

Dong Shu, Yanguang Liu, Huopu Zhang, and Mengnan Du. FinCall-Surprise: A large scale multi-modal benchmark for earning surprise prediction. arXiv:2510.03965; data: github.com/Tizzzy/FinCall-Surprise (Apache-2.0), 2025. URL <https://arxiv.org/abs/2510.03965>.

Nikita Tatarinov et al. Language modeling for the future of finance: A survey into metrics, tasks, and data opportunities. arXiv:2504.07274, 2025. URL <https://arxiv.org/abs/2504.07274>.

Wang et al. AMA-LSTM: Pioneering robust and fair financial audio analysis for stock volatility prediction. NAACL 2024 Industry Track; arXiv:2407.18324, 2024. URL <https://arxiv.org/abs/2407.18324>.

Yutong Yan et al. DatedGPT: Preventing lookahead bias in large language models with time-aware pretraining. arXiv:2603.11838, 2026. URL <https://arxiv.org/abs/2603.11838>.

Linyi Yang, Tin Lok James Ng, Barry Smyth, and Ruihai Dong. HTML: Hierarchical transformer-based multi-task learning for volatility prediction. In *Proceedings of The Web Conference (WWW)*, 2020.

Linyi Yang et al. NumHTML: Numeric-oriented hierarchical transformer model for multi-task financial forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022. URL <https://arxiv.org/abs/2201.01770>.

Ye, Qin, and Xu. Financial risk prediction with multi-round q&a attention network. In *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*, 2020. URL <https://www.ijcai.org/proceedings/2020/631>.

Ding Yu et al. Same company, same signal: The role of identity in earnings call transcripts. arXiv:2412.18029, 2024. URL <https://arxiv.org/abs/2412.18029>.

A Complete Results

The tables below are generated byte-identically by `ecvol report` from the committed result CSVs (`data/results/`) and are the source of truth for the curated tables in §6; the curated tables copy selected cells from them. They span both corpora (FinCall-Surprise, MAEC), the level- v and Δv targets, both the R^2_{OOS} and MSE metrics, and the temporal and ticker-disjoint splits at all horizons.

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
EWMA	0.387*	0.362*	0.145*	-0.219*
HAR-RV	0.464*	0.410*	0.207*	-0.287*
GARCH(1,1)	0.265*	0.193*	-0.140*	-0.541*
GBDT (ticker FE)	0.354*	0.262*	-0.040	-0.839*

Table 7: fincall — temporal split — level- v — R^2_{OOS} vs persistence (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
EWMA	0.357*	0.199*	-0.009	0.046
HAR-RV	0.495*	0.357*	0.211*	0.229*
GARCH(1,1)	0.356*	0.229*	0.088	0.209*
GBDT (ticker FE)	0.412*	0.278*	0.078	0.116*

Table 8: fincall — ticker_disjoint split — level- v — R^2_{OOS} vs persistence (test) (n=484).

A.1 Stage-0/1 baselines (Result Table 1)

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	0.000	0.000	0.000	0.000
EWMA	0.387*	0.362*	0.145*	-0.219*	EWMA	0.398*	0.297*	0.119*	-0.144*
HAR-RV	0.464*	0.410*	0.207*	-0.287*	HAR-RV	0.474*	0.379*	0.311*	0.225*
GARCH(1,1)	0.265*	0.193*	-0.140*	-0.541*	GARCH(1,1)	0.306*	0.213*	0.009	-0.321*
GBDT (ticker FE)	0.344*	0.248*	-0.057	-0.904*	GBDT (ticker FE)	0.394*	0.285*	0.227*	0.233*

Table 9: fincall — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=481).

Table 14: maec — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	1.174	0.419	0.208	0.111
EWMA	0.357*	0.199*	-0.009	0.046	EWMA	0.720*	0.267*	0.178*	0.136*
HAR-RV	0.495*	0.357*	0.211*	0.229*	HAR-RV	0.630*	0.247*	0.165*	0.143*
GARCH(1,1)	0.356*	0.229*	0.088	0.209*	GARCH(1,1)	0.860*	0.335*	0.233*	0.167*
GBDT (ticker FE)	0.407*	0.275*	0.089	0.108	GBDT (ticker FE)	0.758*	0.309*	0.216	0.205*

Table 10: fincall — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=484).

Table 15: fincall — temporal split — level-v — MSE (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	1.142	0.386	0.214	0.233
EWMA	0.444*	0.301*	0.120*	-0.144*	EWMA	0.735*	0.309*	0.216	0.222
HAR-RV	0.507*	0.392*	0.306*	0.206*	HAR-RV	0.577*	0.248*	0.169*	0.180*
GARCH(1,1)	0.387*	0.281*	0.054	-0.309*	GARCH(1,1)	0.735*	0.297*	0.195	0.184*
GBDT (ticker FE)	0.397*	0.333*	0.217*	0.131*	GBDT (ticker FE)	0.672*	0.278*	0.197	0.206*

Table 11: maec — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=507).

Table 16: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	1.174	0.419	0.208	0.111
EWMA	0.398*	0.297*	0.119*	-0.144*	EWMA	0.720*	0.267*	0.178*	0.136*
HAR-RV	0.474*	0.379*	0.311*	0.225*	HAR-RV	0.630*	0.247*	0.165*	0.143*
GARCH(1,1)	0.306*	0.213*	0.009	-0.321*	GARCH(1,1)	0.860*	0.335*	0.233*	0.167*
GBDT (ticker FE)	0.395*	0.306*	0.236*	0.221*	GBDT (ticker FE)	0.771*	0.315*	0.220	0.212*

Table 12: maec — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=514).

Table 17: fincall — temporal split — Δv — MSE (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	1.142	0.386	0.214	0.233
EWMA	0.444*	0.301*	0.120*	-0.144*	EWMA	0.735*	0.309*	0.216	0.222
HAR-RV	0.507*	0.392*	0.306*	0.206*	HAR-RV	0.577*	0.248*	0.169*	0.180*
GARCH(1,1)	0.387*	0.281*	0.054	-0.309*	GARCH(1,1)	0.735*	0.297*	0.195	0.184*
GBDT (ticker FE)	0.407*	0.340*	0.213*	0.186*	GBDT (ticker FE)	0.678*	0.279*	0.195	0.208

Table 13: maec — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=507).

Table 18: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.383	0.571	0.318	0.189
EWMA	0.768*	0.399*	0.280*	0.216*
HAR-RV	0.682*	0.347*	0.221*	0.150*
GARCH(1,1)	0.848*	0.411*	0.301	0.247*
GBDT (ticker FE)	0.834*	0.381*	0.249*	0.164*

Table 19: maec — temporal split — level-v — MSE (test)
(n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.268	0.492	0.302	0.184
EWMA	0.764*	0.346*	0.266*	0.210*
HAR-RV	0.667*	0.306*	0.208*	0.142*
GARCH(1,1)	0.881*	0.384*	0.296	0.242*
GBDT (ticker FE)	0.767*	0.342*	0.230*	0.143*

Table 20: maec — ticker_disjoint split — level-v — MSE
(test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.383	0.571	0.318	0.189
EWMA	0.768*	0.399*	0.280*	0.216*
HAR-RV	0.682*	0.347*	0.221*	0.150*
GARCH(1,1)	0.848*	0.411*	0.301	0.247*
GBDT (ticker FE)	0.820*	0.377*	0.250*	0.154*

Table 21: maec — temporal split — Δv — MSE (test)
(n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.268	0.492	0.302	0.184
EWMA	0.764*	0.346*	0.266*	0.210*
HAR-RV	0.667*	0.306*	0.208*	0.142*
GARCH(1,1)	0.881*	0.384*	0.296	0.242*
GBDT (ticker FE)	0.769*	0.352*	0.233*	0.141*

Table 22: maec — ticker_disjoint split — Δv — MSE (test)
(n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.439*	0.319	0.037	-0.577	Ridge (text)	0.028*	0.030*	0.015	-0.144*
MLP (text)	0.147*	-0.241*	-1.101*	-2.633*	MLP (text)	-0.108*	-0.175*	-0.265*	-1.042
Ridge (past-vol)	0.415*	-1.052	-2.025	-0.371*	Ridge (past-vol)	0.415*	-1.053	-2.858	-0.667
MLP (past-vol)	0.456*	-0.228	-2.706	-2.477	MLP (past-vol)	0.390	-1.017	-2.249	-0.760
Ridge (text+vol)	0.448*	0.328	0.045	-1.091	Ridge (text+vol)	-1.678	-3.076	-3.586	-0.468
MLP (text+vol)	-2.092	-7.189	-18.384	-26.901	MLP (text+vol)	-1.186	-1.571	-2.112	-1.295

Table 23: fincall — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=481).

Table 25: fincall — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.416	0.194*	-0.047*	0.112	Ridge (text)	-0.044*	-0.026*	0.008	0.171
MLP (text)	-0.362*	-1.801*	-3.485*	-2.872*	MLP (text)	-0.195*	-0.271*	-0.339*	-0.204*
Ridge (past-vol)	0.432	0.306	0.191*	0.224*	Ridge (past-vol)	0.451	0.306	0.191*	0.224*
MLP (past-vol)	0.482*	0.365*	0.243*	0.212*	MLP (past-vol)	0.492*	0.363*	0.230*	0.249*
Ridge (text+vol)	0.418	0.211	-0.006	0.139	Ridge (text+vol)	0.223*	0.129*	0.094	0.236
MLP (text+vol)	-0.254*	-1.556*	-3.005*	-2.486*	MLP (text+vol)	0.275*	0.140*	-0.061*	0.051

Table 24: fincall — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=484).

Table 26: fincall — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=484).

A.2 Stage-2 text heads (Result Table 2)

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.478*	0.225*	-0.033*	-0.322*
MLP (text)	-0.748*	-1.753*	-3.272*	-6.016*
Ridge (past-vol)	0.437	0.222*	0.168	0.208
MLP (past-vol)	0.472*	0.375	0.301*	0.247*
Ridge (text+vol)	0.482*	0.279	0.132	-0.013*
MLP (text+vol)	-0.405*	-1.460*	-2.977*	-5.058*

Table 27: maec — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.466*	0.277	0.117*	-0.036*
MLP (text)	-0.115*	-0.956*	-1.583*	-2.635*
Ridge (past-vol)	0.455*	0.329	0.261	0.262
MLP (past-vol)	0.482*	0.369*	0.281	0.250
Ridge (text+vol)	0.451	0.299	0.186	0.090*
MLP (text+vol)	-0.077*	-0.884*	-1.418*	-2.992*

Table 28: maec — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.017*	0.013*	0.047*	-0.074*
MLP (text)	-0.301*	-0.164*	-0.338*	-0.511*
Ridge (past-vol)	0.455	0.313	0.244	0.212
MLP (past-vol)	0.497*	0.395*	0.323*	0.261*
Ridge (text+vol)	-0.005*	-0.142*	0.211	0.138
MLP (text+vol)	0.259*	0.242*	0.068*	-0.080*

Table 29: maec — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.049*	0.056*	0.048*	0.047*
MLP (text)	-0.051*	-0.108*	-0.165*	-0.199*
Ridge (past-vol)	0.455*	0.329	0.261	0.262
MLP (past-vol)	0.489*	0.389*	0.328*	0.294*
Ridge (text+vol)	0.330	0.142*	0.204	0.163
MLP (text+vol)	0.366	0.244	0.120*	-0.111*

Table 30: maec — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.659*	0.285	0.200	0.176
MLP (text)	1.001*	0.519*	0.437*	0.405*
Ridge (past-vol)	0.687*	0.859	0.629	0.153*
MLP (past-vol)	0.639*	0.514	0.771	0.387
Ridge (text+vol)	0.648*	0.281	0.199	0.233
MLP (text+vol)	3.631	3.428	4.032	3.109

Table 31: fincall — temporal split — level-v — MSE (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.667	0.311*	0.224*	0.207
MLP (text)	1.556*	1.080*	0.958*	0.903*
Ridge (past-vol)	0.649	0.268	0.173*	0.181*
MLP (past-vol)	0.592*	0.245*	0.162*	0.184*
Ridge (text+vol)	0.665	0.304	0.215	0.201
MLP (text+vol)	1.432*	0.986*	0.856*	0.813*

Table 32: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	1.141*	0.406*	0.205	0.127*
MLP (text)	1.302*	0.492*	0.263*	0.228
Ridge (past-vol)	0.687*	0.860	0.803	0.186
MLP (past-vol)	0.717	0.844	0.676	0.196
Ridge (text+vol)	3.145	1.706	0.954	0.164
MLP (text+vol)	2.567	1.076	0.647	0.256

Table 33: fincall — temporal split — Δv — MSE (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	1.193*	0.396*	0.212	0.193
MLP (text)	1.365*	0.490*	0.286*	0.281*
Ridge (past-vol)	0.627	0.268	0.173*	0.181*
MLP (past-vol)	0.580*	0.246*	0.164*	0.175*
Ridge (text+vol)	0.888*	0.336*	0.194	0.178
MLP (text+vol)	0.828*	0.332*	0.227*	0.221

Table 34: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.722*	0.443*	0.329*	0.250*
MLP (text)	2.416*	1.573*	1.359*	1.325*
Ridge (past-vol)	0.778	0.445*	0.265	0.150
MLP (past-vol)	0.730*	0.357	0.222*	0.142*
Ridge (text+vol)	0.716*	0.412	0.276	0.191*
MLP (text+vol)	1.943*	1.405*	1.266*	1.144*

Table 35: maec — temporal split — level-v — MSE (test) (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.677*	0.356	0.266*	0.190*
MLP (text)	1.413*	0.963*	0.779*	0.668*
Ridge (past-vol)	0.691*	0.330	0.223	0.136
MLP (past-vol)	0.656*	0.311*	0.217	0.138
Ridge (text+vol)	0.695	0.345	0.246	0.167*
MLP (text+vol)	1.365*	0.927*	0.729*	0.734*

Table 36: maec — ticker_disjoint split — level-v — MSE (test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	1.358*	0.564*	0.303*	0.203*
MLP (text)	1.798*	0.665*	0.426*	0.285*
Ridge (past-vol)	0.753	0.392	0.241	0.149
MLP (past-vol)	0.695*	0.346*	0.215*	0.140*
Ridge (text+vol)	1.390*	0.653*	0.251	0.163
MLP (text+vol)	1.025*	0.433*	0.297*	0.204*

Table 37: maec — temporal split — Δv — MSE (test) (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	1.206*	0.464*	0.287*	0.175*
MLP (text)	1.332*	0.546*	0.351*	0.220*
Ridge (past-vol)	0.691*	0.330	0.223	0.136
MLP (past-vol)	0.648*	0.301*	0.203*	0.130*
Ridge (text+vol)	0.850	0.422*	0.240	0.154
MLP (text+vol)	0.804	0.372	0.265*	0.204*

Table 38: maec — ticker_disjoint split — Δv — MSE (test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	0.362	0.126*	-0.338*	-1.109*
mlp eGeMAPS	-9.446	-22.924	-39.453	-63.845
ridge eGeMAPS +vol	0.347	0.258	-0.046	-0.668
mlp eGeMAPS +vol	-9.226	-21.289	-46.203	-77.488
ridge WavLM	0.359	0.122*	-0.427*	-1.167*
mlp WavLM	-0.117*	-0.965*	-2.387*	-5.448*
ridge WavLM +vol	0.343	0.245	-0.390	-1.599
mlp WavLM +vol	-4.487	-14.363	-32.806	-59.317
ridge emotion2vec+	0.353	0.138*	-0.289*	-1.025*
mlp emotion2vec+	-0.385*	-1.880*	-4.314*	-7.784*
ridge emotion2vec+ +vol	0.401	-0.583	-1.139	-0.422*
mlp emotion2vec+ +vol	-6.056	-20.286	-41.247	-64.876
ridge WavLM+eGeMAPS	0.358	0.116*	-0.379*	-1.172*
mlp WavLM+eGeMAPS	-2.476	-6.938	-11.288	-17.515
ridge WavLM+eGeMAPS +vol	0.338	0.221	-0.139	-1.661
mlp WavLM+eGeMAPS +vol	-5.037	-14.919	-30.138	-43.263
ridge WavLM+text +vol	0.443*	0.321	0.014	-1.142
mlp WavLM+text +vol	-2.902	-8.912	-22.030	-33.025

Table 39: fincall — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	0.410	0.106*	-0.238*	-0.106*
mlp eGeMAPS	-0.074	-1.156*	-1.969*	-1.835*
ridge eGeMAPS +vol	0.416	0.270	0.164*	0.164*
mlp eGeMAPS +vol	-0.338	-1.497	-2.709	-2.609
ridge WavLM	0.397	0.092*	-0.235*	-0.092*
mlp WavLM	-0.355*	-1.902*	-4.384*	-4.025*
ridge WavLM +vol	0.403	0.137*	0.088	0.137*
mlp WavLM +vol	-0.320*	-1.388*	-2.985*	-3.161*
ridge emotion2vec+	0.409	0.125*	-0.202*	-0.061*
mlp emotion2vec+	-1.476	-5.431	-10.233	-9.841
ridge emotion2vec+ +vol	0.416	0.298	0.195*	0.219*
mlp emotion2vec+ +vol	-1.243	-4.109	-8.681	-7.141
ridge WavLM+eGeMAPS	0.394	0.094*	-0.230*	-0.094*
mlp WavLM+eGeMAPS	-0.308*	-1.393*	-2.760*	-2.331*
ridge WavLM+eGeMAPS +vol	0.400	0.137*	0.073	0.137*
mlp WavLM+eGeMAPS +vol	-0.302*	-1.323*	-2.551*	-2.151*
ridge WavLM+text +vol	0.412	0.205	0.001	0.137*
mlp WavLM+text +vol	-0.136*	-1.278*	-2.540*	-2.041*

Table 40: fincall — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=484).

A.3 Stage-3 audio heads (Result Table 3)

FinCall-Surprise only (MAEC ships no raw audio).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	-0.000*	-0.007*	-0.015	-0.000*
mlp eGeMAPS	-0.492*	-0.218*	-0.272*	-0.335*
ridge eGeMAPS +vol	0.219	-1.834	-4.099	-0.611
mlp eGeMAPS +vol	0.187	-1.680	-5.488	-5.341
ridge WavLM	-0.005*	-0.007*	-0.045	-0.035*
mlp WavLM	-0.176*	-0.248*	-0.449*	-0.911
ridge WavLM +vol	0.265	-1.026	-2.032	-0.511
mlp WavLM +vol	-1.219	-2.099	-2.908	-2.441
ridge emotion2vec+	-0.018*	-0.028*	-0.056	-0.061*
mlp emotion2vec+	-0.096*	-0.256*	-0.468*	-1.241
ridge emotion2vec+ +vol	0.229	-2.052	-3.031	-0.611
mlp emotion2vec+ +vol	-1.692	-3.466	-3.611	-2.311
ridge WavLM+eGeMAPS	-0.005*	0.001*	-0.015	-0.025*
mlp WavLM+eGeMAPS	-0.296*	-0.497*	-0.641*	-1.341
ridge WavLM+eGeMAPS +vol	0.214	-1.099	-3.706	-0.511
mlp WavLM+eGeMAPS +vol	-1.810	-3.234	-3.991	-3.111
ridge WavLM+text +vol	-1.697	-3.043	-0.323	-0.411
mlp WavLM+text +vol	-1.456	-2.193	-3.917	-3.111

Table 41: fincall — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=481).

Model	$\tau=3$	$\tau=7$	Model	$\tau=30$	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	-0.011*	-0.013*	ridge eGeMAPS	0.002	0.674	0.345*	0.265*	0.258*
mlp eGeMAPS	-0.167*	-0.231*	mlp eGeMAPS	-0.171*	1.227	0.832*	0.634*	0.666*
ridge eGeMAPS +vol	0.415	0.253	ridge eGeMAPS +vol	0.156	0.667	0.281	0.179*	0.188
mlp eGeMAPS +vol	0.353	0.207	mlp eGeMAPS +vol	0.170	1.529	0.963	0.792	0.843
ridge WavLM	-0.038*	-0.035*	ridge WavLM	-0.027*	0.689	0.350*	0.264*	0.256*
mlp WavLM	-0.248*	-0.323*	mlp WavLM	-0.351*	1.547*	1.119*	1.150*	1.172*
ridge WavLM +vol	0.337*	0.171*	ridge WavLM +vol	0.174*	0.682	0.333*	0.195	0.198
mlp WavLM +vol	0.160*	0.009*	mlp WavLM +vol	-0.174*	1.508*	0.921*	0.852*	0.971*
ridge emotion2vec+	-0.002*	-0.003*	ridge emotion2vec+	0.016	0.675	0.338*	0.257*	0.249*
mlp emotion2vec+	-0.257*	-0.397*	mlp emotion2vec+	-0.492*	2.828	2.480	2.400	2.539
ridge emotion2vec+ +vol	0.435	0.297	ridge emotion2vec+ +vol	0.198*	0.667	0.271	0.172*	0.182*
mlp emotion2vec+ +vol	0.105*	-0.139*	mlp emotion2vec+ +vol	-0.255*	2.562	1.970	2.068	1.909
ridge WavLM+eGeMAPS	-0.050*	-0.042*	ridge WavLM+eGeMAPS	-0.026*	0.692	0.350*	0.263*	0.256*
mlp WavLM+eGeMAPS	-0.225*	-0.366*	mlp WavLM+eGeMAPS	-0.451*	1.493*	0.923*	0.803*	0.778*
ridge WavLM+eGeMAPS +vol	0.304*	0.147*	ridge WavLM+eGeMAPS +vol	0.144	0.686	0.333*	0.198	0.200
mlp WavLM+eGeMAPS +vol	0.109*	-0.004*	mlp WavLM+eGeMAPS +vol	-0.111*	1.487*	0.896*	0.759*	0.736*
ridge WavLM+text +vol	0.159*	0.072*	ridge WavLM+text +vol	0.033	0.671	0.307	0.214	0.199
mlp WavLM+text +vol	0.228*	0.096*	mlp WavLM+text +vol	-0.084*	1.298*	0.878*	0.756*	0.709*

Table 42: fincall — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=484).

Table 44: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	$\tau=3$	$\tau=7$	Model	$\tau=30$	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	0.750	0.366*	ridge eGeMAPS	0.278*	1.175*	0.421*	0.211	0.112*
mlp eGeMAPS	12.267	10.015	mlp eGeMAPS	8.416	1.752*	0.510*	0.265*	0.149*
ridge eGeMAPS +vol	0.767	0.311	ridge eGeMAPS +vol	0.218*	0.917	1.186	1.061	0.182
mlp eGeMAPS +vol	12.008	9.331	mlp eGeMAPS +vol	9.819	0.954	1.122	1.350	0.704
ridge WavLM	0.753	0.368*	ridge WavLM	0.297*	1.180*	0.422*	0.217	0.115*
mlp WavLM	1.312*	0.823*	mlp WavLM	0.704*	1.381*	0.523*	0.301*	0.221
ridge WavLM +vol	0.772	0.316	ridge WavLM +vol	0.289*	0.863	0.848	0.631	0.174
mlp WavLM +vol	6.443	6.432	mlp WavLM +vol	7.032	2.606	1.297	0.813	0.389
ridge emotion2vec+	0.759	0.361*	ridge emotion2vec+	0.226*	1.196*	0.431*	0.220	0.118*
mlp emotion2vec+	1.626*	1.206*	mlp emotion2vec+	0.979*	1.288*	0.526*	0.305*	0.250*
ridge emotion2vec+ +vol	0.703	0.663	ridge emotion2vec+ +vol	0.445*	0.905	1.278	0.838	0.187
mlp emotion2vec+ +vol	8.286	8.911	mlp emotion2vec+ +vol	7.881	3.161	1.870	0.959	0.370
ridge WavLM+eGeMAPS	0.754	0.370*	ridge WavLM+eGeMAPS	0.287*	1.180*	0.418*	0.211	0.115*
mlp WavLM+eGeMAPS	4.082	3.323	mlp WavLM+eGeMAPS	2.556	1.522*	0.627*	0.341*	0.261*
ridge WavLM+eGeMAPS +vol	0.777	0.326	ridge WavLM+eGeMAPS +vol	0.237*	0.923	0.879	0.979	0.172
mlp WavLM+eGeMAPS +vol	7.089	6.664	mlp WavLM+eGeMAPS +vol	6.477	3.300	1.773	1.038	0.466
ridge WavLM+text +vol	0.654*	0.284	ridge WavLM+text +vol	0.205*	3.167	1.692	0.275	0.161
mlp WavLM+text +vol	4.582	4.150	mlp WavLM+text +vol	4.791	2.884	1.337	1.023	0.458

Table 43: fincall — temporal split — level-v — MSE (test) (n=481).

Table 45: fincall — temporal split — Δv — MSE (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	1.155*	0.391*	0.214	0.233
mlp eGeMAPS	1.333*	0.475*	0.251*	0.258*
ridge eGeMAPS +vol	0.668	0.288	0.180	0.186*
mlp eGeMAPS +vol	0.739	0.306	0.199	0.217
ridge WavLM	1.185*	0.399*	0.219*	0.228
mlp WavLM	1.426*	0.510*	0.289*	0.275*
ridge WavLM +vol	0.758*	0.320*	0.190	0.184*
mlp WavLM +vol	0.959*	0.382*	0.251*	0.261*
ridge emotion2vec+	1.144*	0.387*	0.211	0.233
mlp emotion2vec+	1.436*	0.539*	0.319*	0.335*
ridge emotion2vec+ +vol	0.645	0.271	0.171*	0.182*
mlp emotion2vec+ +vol	1.022*	0.439*	0.268*	0.275*
ridge WavLM+eGeMAPS	1.199*	0.402*	0.219*	0.227
mlp WavLM+eGeMAPS	1.400*	0.527*	0.312*	0.316*
ridge WavLM+eGeMAPS +vol	0.795*	0.329*	0.204	0.184*
mlp WavLM+eGeMAPS +vol	1.017*	0.387*	0.239*	0.260*
ridge WavLM+text +vol	0.961*	0.358*	0.207	0.186
mlp WavLM+text +vol	0.882*	0.348*	0.232*	0.216

Table 46: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.464	0.410	0.207	-0.287	HAR-RV	0.464	0.410	0.207	-0.287
GBDT (ticker FE)	0.354	0.262	-0.040	-0.839	GBDT (ticker FE)	0.344	0.248	-0.057	-0.904
Text (BGE+vol)	0.448	0.328	0.045	-1.091	Text (BGE+vol)	-1.678	-3.076	-3.586	-0.468
Audio (WavLM+eG+vol)	0.338	0.221	-0.139	-1.661	Audio (WavLM+eG+vol)	0.214	-1.099	-3.706	-0.548
Fusion (stack)	0.448	0.370	0.125	-1.045	Fusion (stack)	0.232	-0.326	-1.828	-0.010*

Table 47: fincall — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=481).

Table 49: fincall — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=481).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.495	0.357	0.211	0.229	HAR-RV	0.495	0.357	0.211	0.229
GBDT (ticker FE)	0.412	0.278	0.078	0.116	GBDT (ticker FE)	0.407	0.275	0.089	0.108
Text (BGE+vol)	0.418	0.211	-0.006	0.132	Text (BGE+vol)	0.223*	0.129*	0.094	0.236
Audio (WavLM+eG+vol)	0.400	0.137	0.073	0.144	Audio (WavLM+eG+vol)	0.304*	0.147*	0.044	0.209
Fusion (stack)	0.450	0.264	0.050	0.171	Fusion (stack)	0.436	0.292	0.133	0.244*

Table 48: fincall — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=484).

Table 50: fincall — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=484).

A.4 Consolidated ablation grid (Result Table 4)

One pre-registered canonical model per stage; * marks Holm-corrected significance vs. Stage-1 within the Δv confirmatory family (two-sided—read direction from the value).

On the absence of a Result Table 5. The Stage-5 LLM feature extraction was blocked at its pre-registered validity gate (§6.6, Table 6); no corpus-scale LLM features were produced, by design.

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.507	0.392	0.306	0.206
GBDT (ticker FE)	0.397	0.333	0.217	0.131
Text (BGE+vol)	0.482	0.279	0.132	-0.013
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 51: maec — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=507).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.474	0.379	0.311	0.225
GBDT (ticker FE)	0.395	0.306	0.236	0.221
Text (BGE+vol)	0.451	0.299	0.186	0.090
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 52: maec — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=514).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.507	0.392	0.306	0.206
GBDT (ticker FE)	0.407	0.340	0.213	0.186
Text (BGE+vol)	-0.005*	-0.142*	0.211	0.138
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 53: maec — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=507).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.474	0.379	0.311	0.225
GBDT (ticker FE)	0.394	0.285	0.227	0.233
Text (BGE+vol)	0.330	0.142*	0.204	0.163
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 54: maec — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=514).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.174	0.419	0.208	0.111
HAR-RV	0.630	0.247	0.165	0.143
GBDT (ticker FE)	0.758	0.309	0.216	0.205
Text (BGE+vol)	0.648	0.281	0.199	0.233
Audio (WavLM+eG+vol)	0.777	0.326	0.237	0.296
Fusion (stack)	0.649	0.264	0.182	0.225

Table 55: fincall — temporal split — level-v — MSE (test) (n=481).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.142	0.386	0.214	0.233
HAR-RV	0.577	0.248	0.169	0.180
GBDT (ticker FE)	0.672	0.278	0.197	0.206
Text (BGE+vol)	0.665	0.304	0.215	0.201
Audio (WavLM+eG+vol)	0.686	0.333	0.198	0.200
Fusion (stack)	0.628	0.284	0.203	0.193

Table 56: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.174	0.419	0.208	0.111
HAR-RV	0.630	0.247	0.165	0.143
GBDT (ticker FE)	0.771	0.315	0.220	0.212
Text (BGE+vol)	3.145	1.706	0.954	0.164
Audio (WavLM+eG+vol)	0.923	0.879	0.979	0.172
Fusion (stack)	0.902	0.555	0.588	0.113*

Table 57: fincall — temporal split — Δv — MSE (test) (n=481).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.142	0.386	0.214	0.233
HAR-RV	0.577	0.248	0.169	0.180
GBDT (ticker FE)	0.678	0.279	0.195	0.208
Text (BGE+vol)	0.888*	0.336*	0.194	0.178
Audio (WavLM+eG+vol)	0.795*	0.329*	0.204	0.184
Fusion (stack)	0.644	0.273	0.185	0.176*

Table 58: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.383	0.571	0.318	0.189
HAR-RV	0.682	0.347	0.221	0.150
GBDT (ticker FE)	0.834	0.381	0.249	0.164
Text (BGE+vol)	0.716	0.412	0.276	0.191
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 59: maec — temporal split — level-v — MSE (test) (n=507).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.268	0.492	0.302	0.184
HAR-RV	0.667	0.306	0.208	0.142
GBDT (ticker FE)	0.767	0.342	0.230	0.143
Text (BGE+vol)	0.695	0.345	0.246	0.167
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 60: maec — ticker_disjoint split — level-v — MSE (test) (n=514).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.383	0.571	0.318	0.189
HAR-RV	0.682	0.347	0.221	0.150
GBDT (ticker FE)	0.820	0.377	0.250	0.154
Text (BGE+vol)	1.390*	0.653*	0.251	0.163
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 61: maec — temporal split — Δv — MSE (test) (n=507).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.268	0.492	0.302	0.184
HAR-RV	0.667	0.306	0.208	0.142
GBDT (ticker FE)	0.769	0.352	0.233	0.141
Text (BGE+vol)	0.850	0.422*	0.240	0.154
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 62: maec — ticker_disjoint split — Δv — MSE (test) (n=514).